

Harshal Abdulla

Associate / graduate software engineer · backend-leaning full stack

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SELECTED PROJECTS

Restaurant ordering and till system

Commercial, sole developer · 2026

Live in a restaurant in Co. Kildare, in use every service for over five weeks

- Built four surfaces (customer kiosk, kitchen till, table service, admin) from one Expo codebase against one Postgres database, in seven weeks, as the only developer. 154 commits, 15,109 lines of TypeScript.
- Put pricing and entitlement rules in plpgsql behind security definer so all four surfaces get the same answer and no client can bypass the till: around 70 Postgres functions behind row level security across 94 migrations.
- Removed the application server entirely. Static hosting with the client talking to Postgres through PostgREST, and row level security as the security boundary.
- Money is integer euro cents with no floating point. Order lines are immutable snapshots and corrections are soft deletes, so history is never rewritten.
- Wrote a property-based test suite from scratch with no framework and no dependencies: 43 properties, 560,015 generated cases per run in about a second, seeded so any failure replays exactly. Used it to measure an order-dependent pricing effect across 20,000 baskets and lock the guarantee it rests on into a test.

Stack: TypeScript 5.9, React 19.2, React Native, Expo SDK 54, Zustand, PostgreSQL via Supabase, row level security, pg_cron · Demo: [order-dependent-pricing.surge.sh](#)

Fault-tolerant notification system

Personal · in progress

- Producer writes a PENDING row to Postgres *before* publishing to Kafka, so a crash between the two leaves a durable record a recovery sweep picks up.
- Consumer reads row status before acting, which makes it idempotent by design and absorbs Kafka's at-least-once duplicates as a no-op. Retry and dead-letter path designed, not yet built.

Stack: Python, Apache Kafka, PostgreSQL, Redis, Docker Compose · github.com/Harshal-Abdulla/Kafka-Notification-System

Twix

Group MSc project, team of five · December 2024

- Led a team of five building a Twitter-style microblog: search a user, follow them, post tweets, read a feed. My own share was the interface and the testing.
- Followers are notified through an observer-pattern implementation in the backend; the feed polls the REST API every three seconds.

Stack: Java 21, Spring Boot 3.4, PostgreSQL, Spring Data JPA, React, TypeScript, Vite, Axios, Bootstrap · github.com/Harshal-Abdulla/twix

EXPERIENCE

Frontend Engineer, Cofount Labs LLP

September 2023 – August 2024 · Remote

- Worked from Figma designs, maintaining existing client websites and building new features into them. Client names are under NDA.

EDUCATION

MSc Computer Science (Software Engineering)

2024 – 2025

Maynooth University, Ireland · First Class Honours

Software testing and quality assurance, distributed systems, advanced object-oriented programming and scalable systems.

B.Tech Computer Science and Technology

2019 – 2023

Jyothi Engineering College, Kerala, India

Microsoft Certified: Azure Fundamentals (AZ-900), March 2026

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, SQL, Java
Frontend	React, React Native, Next.js, Expo, Zustand, Tailwind CSS
Backend & data	Node.js, Express, Spring Boot, PostgreSQL, Supabase, MongoDB, Redis, REST
Tooling & testing	Apache Kafka, Docker Compose, GitHub Actions, Git, property-based testing, row level security policy testing